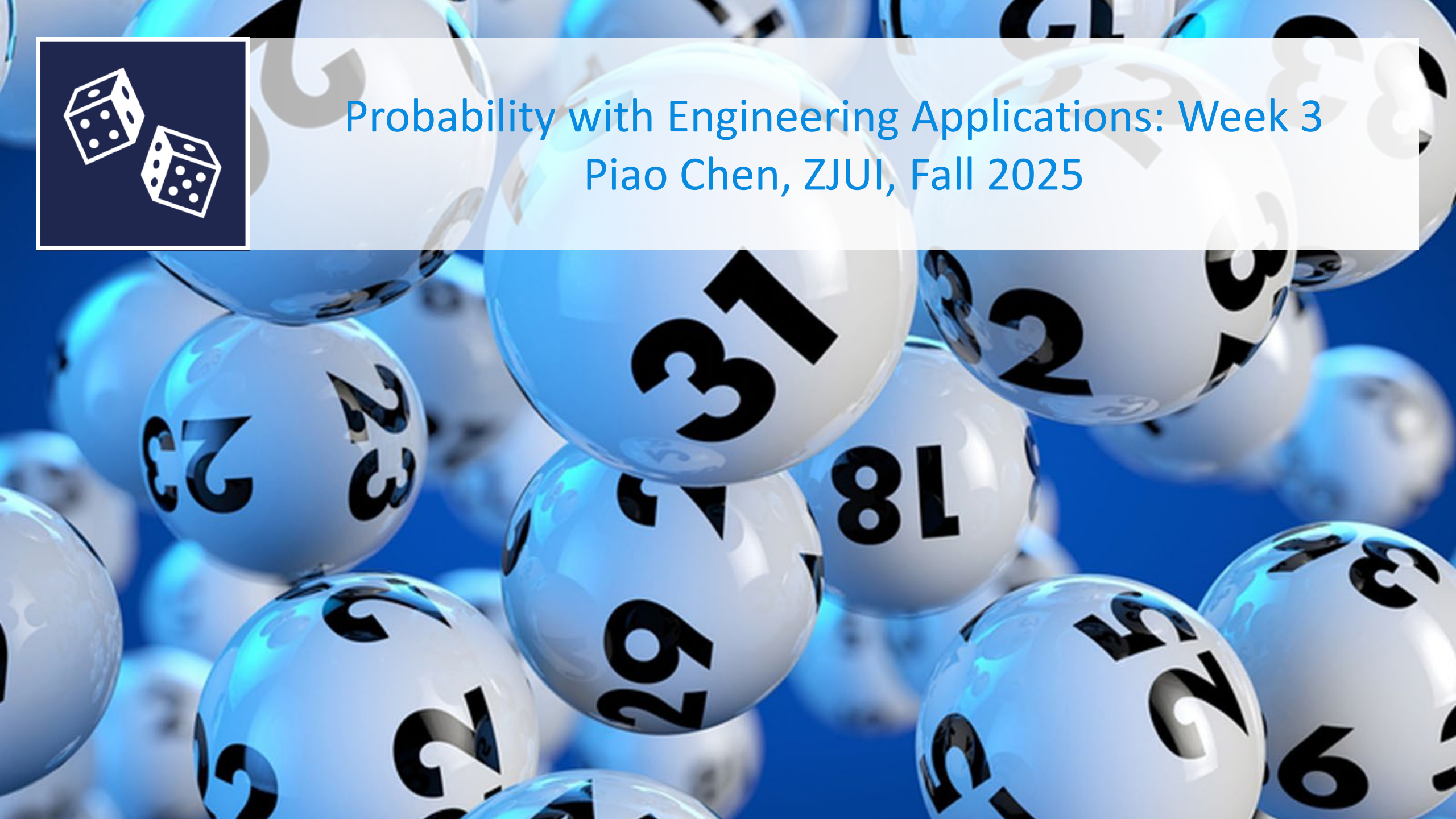




Probability with Engineering Applications: Week 3

Piao Chen, ZJUI, Fall 2025



Recap of previous lectures

- Sample space, event, probability function
- Sum and complement rules of probability functions
- Conditional probabilities
- Law of total probability and Bayes' rule
- Independence of events

Programme

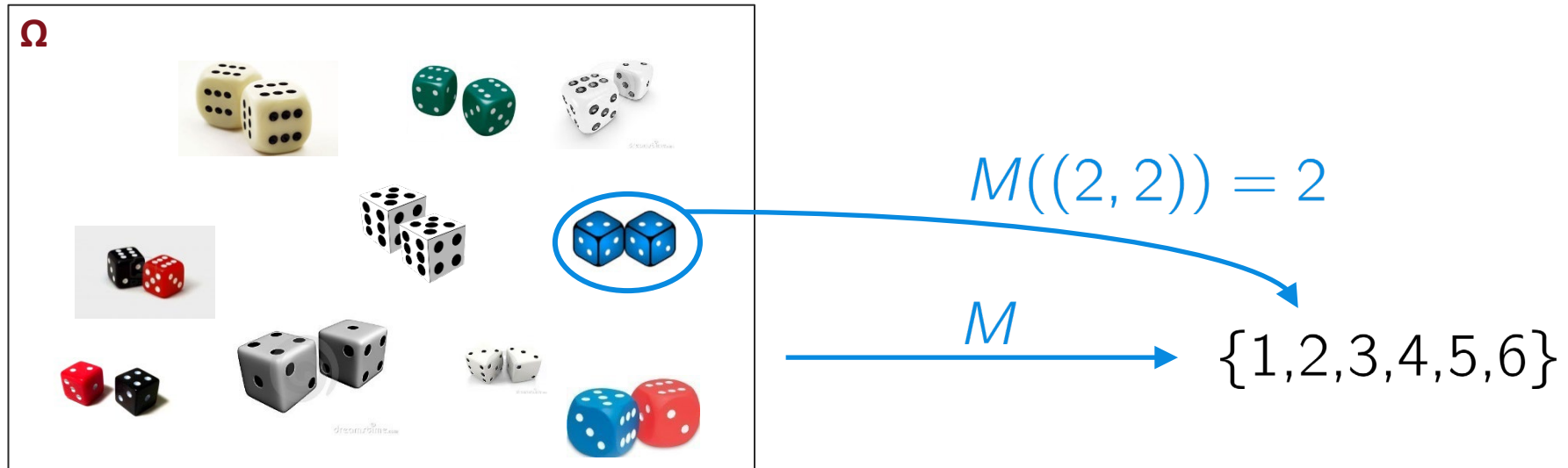
- Discrete random variables
- Discrete distributions

Discrete random variables

Discrete random variable: Motivation

Throw two dice, and let M be the maximum.

$$\Omega = \{(1, 1), (1, 2), \dots, (6, 5), (6, 6)\}$$



Discrete random variable

Definition:

Let Ω be a sample space. A discrete random variable is a function $X : \Omega \rightarrow \mathbb{R}$ that takes on a finite number of values $a_1, a_2, \dots, a_n$ or an infinite number of values $a_1, a_2, a_3, \dots$.



The probability mass function

Definition:

The probability mass function p of a discrete random variable X is the function $p : \mathbb{R} \rightarrow [0, 1]$ defined by

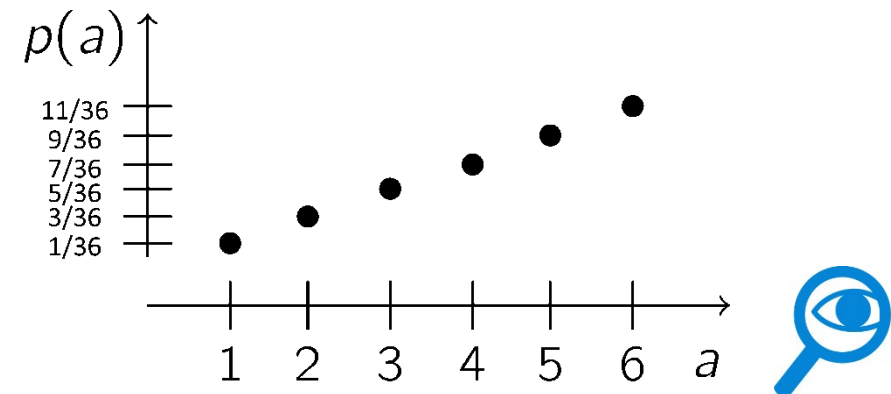
$$p(a) = P(X = a) \quad \text{for} \quad -\infty < a < \infty.$$

Note that $p(a) = 0$ for most values of a .

Example:

Probability mass function of maximum of 2 dice

a	1	2	3	4	5	6
$p(a)$	1/36	3/36	5/36	7/36	9/36	11/36



The probability mass function of the maximum

Ω

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$$P(\{(i,j)\}) = \frac{1}{36} \text{ for any } (i,j)$$

$$p(1) = \frac{1}{36}$$

$$p(2) = \frac{3}{36}$$

$$p(3) = \frac{5}{36}$$

$$p(4) = \frac{7}{36}$$

$$p(5) = \frac{9}{36}$$

$$p(6) = \frac{11}{36}$$

Random Variables Example 1

Let X denote the random variable that is defined as the sum of two fair dice; then

Random Variables Example 2

- Suppose that our experiment consists of **tossing two fair coins**. Letting Y denote **the number of heads** appearing, then

Random Variables Example 3

- Suppose that we **toss a coin until the first head appears**
- Assume a **probability p of coming up heads**, on each flip.
- Letting **N** denote the **number of flips required**, and assume that the outcome of successive flips are independent

Random Variables Example 4

- Suppose that our experiment consists of seeing how long a commodity smart phone can operate before failing.
- Suppose also that we are not primarily interested in the actual lifetime of the phone but only if the phone lasts at least two years.
- We can define the random variable I by

$$I = \begin{cases} 1, & \text{if the lifetime of battery is two or more years} \\ 0, & \text{otherwise} \end{cases}$$

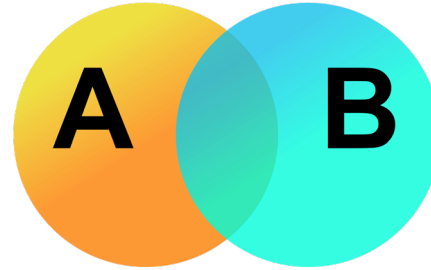
- If E denotes the event that the phone lasts two or more years, then the random variable I is known as the *indicator random variable for event E* . (Note that I equals 1 or 0 depending on whether or not E occurs.)

Random Variables Example 5

- Suppose that **independent trials**, each of which results in any of **m possible outcomes** with respective **probabilities** $p_1, \dots, p_m$, $\sum_{i=1}^m p_i = 1$ are continually performed. Let X denote the number of trials needed until each outcome has occurred at least once.

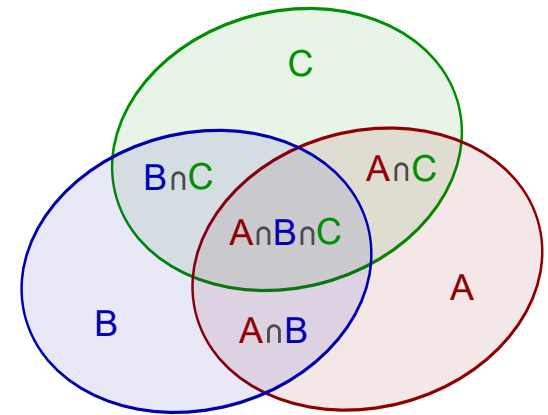
Review: Inclusion-Exclusion Principle

For two events:



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For three events:



$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

The distribution function

Definition:

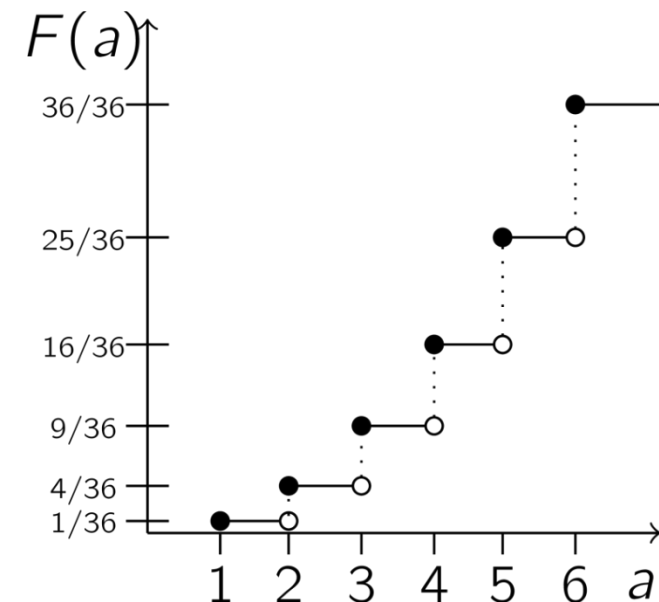
The distribution function F of a discrete random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(a) = P(X \leq a) \quad \text{for} \quad -\infty < a < \infty.$$

Example:

Distribution function of the maximum of 2 dice.

a	1	2	3	4	5	6
$p(a)$	$1/36$	$3/36$	$5/36$	$7/36$	$9/36$	$11/36$



Properties of a distribution function

1. For $a \leq b$ one has that $F(a) \leq F(b)$. This property is an immediate consequence of the fact that $a \leq b$ implies that the event $\{X \leq a\}$ is contained in the event $\{X \leq b\}$.
2. Since $F(a)$ is a probability, the value of the distribution function is always between 0 and 1. Moreover,

$$\begin{aligned}\lim_{a \rightarrow +\infty} F(a) &= \lim_{a \rightarrow +\infty} P(X \leq a) = 1 \\ \lim_{a \rightarrow -\infty} F(a) &= \lim_{a \rightarrow -\infty} P(X \leq a) = 0.\end{aligned}$$

3. F is right-continuous, i.e., one has

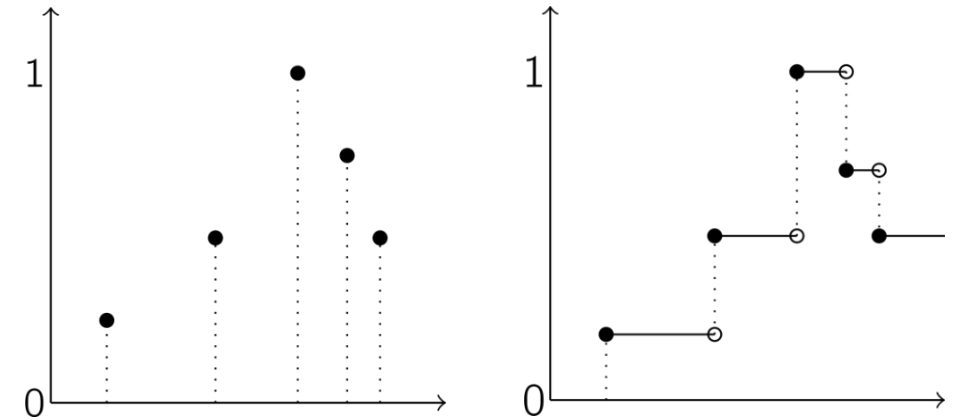
$$\lim_{\varepsilon \downarrow 0} F(a + \varepsilon) = F(a).$$



Probability mass and distribution function

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Consider the two figures on the right.
Determine whether they show a probability mass function (pmf), a distribution function or neither.



- A.** Left: pmf; Right: distribution function
- B.** Left: neither; Right: distribution function
- C.** Left: pmf; Right: neither
- D.** Left: neither; Right: neither

Probability mass and distribution function

$$p(x) = \begin{cases} \frac{1}{10}(3x - 2), & x = 0, 1, 2, 3 \\ 0, & \text{otherwise} \end{cases} \quad F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

- A. Left: pmf; Right: distribution function
- B. Left: neither; Right: distribution function
- C. Left: pmf; Right: neither
- D. Left: neither; Right: neither

Discrete distributions

Context: transmitting a message with errors

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Suppose a source transmits a message consisting of bits. With probability p , a bit is received correctly. Hence with probability $1 - p$, it is incorrectly received. The message consists of 10 bits and $p = 0.95$.



What is the probability the message contains exactly 2 errors?

As soon as the source starts transmitting you check whether a bit containing an error is sent.

What is the probability that the fifth bit sent is the first bit with an error?



Bernoulli distribution

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Definition:

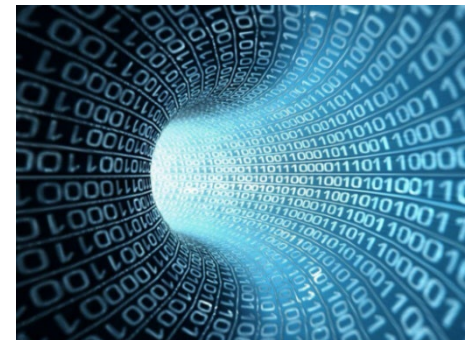
A random variable X has a Bernoulli distribution if X only takes on the values 0 and 1, with probabilities

$$P(X = 1) = p,$$

$$P(X = 0) = 1 - p.$$

Notation: $X \sim \text{Ber}(p)$.

Suppose a source transmits a message consisting of bits. With probability p , a bit is received correctly. Hence with probability $1 - p$, it is received incorrectly.



Bernoulli distribution

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In general:

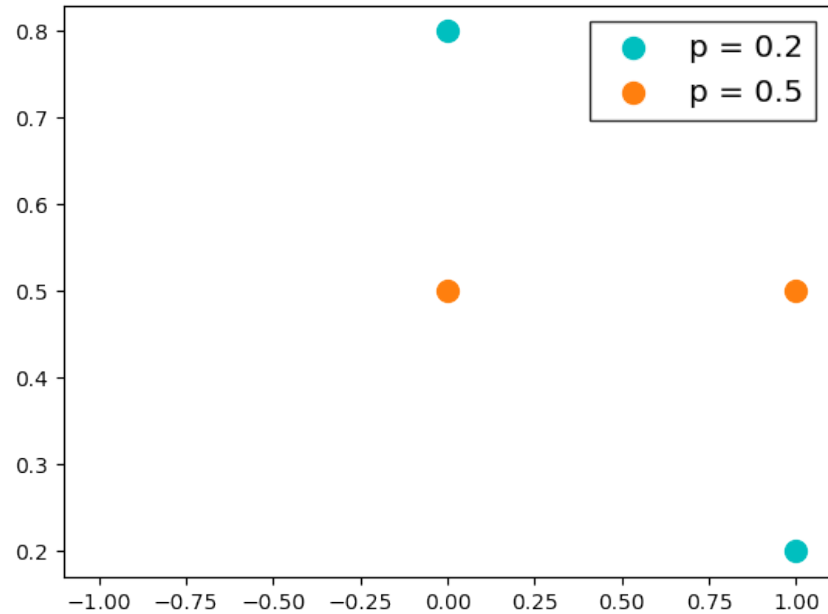
Suppose you conduct an experiment where the outcome is either a success or not. With probability p we observe a success. Then we speak of a Bernoulli distribution with parameter p .



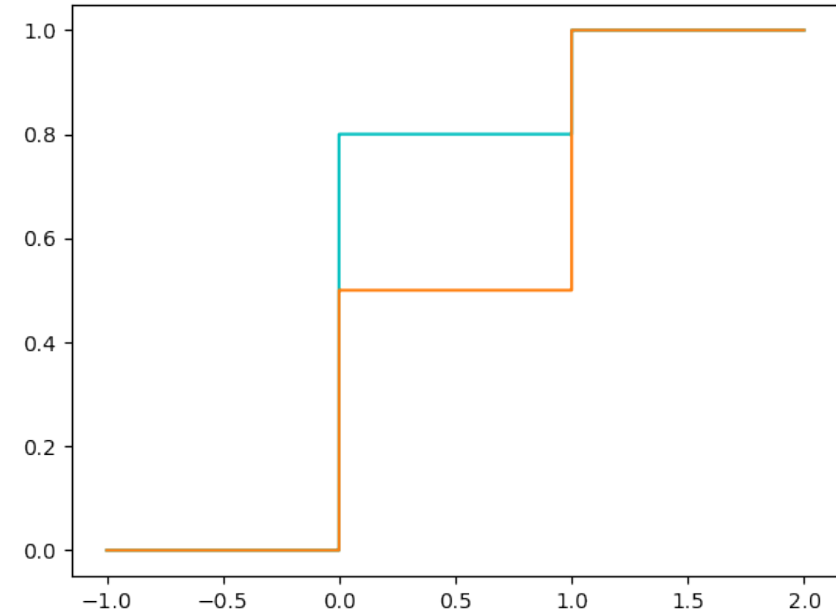
Bernoulli distribution

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Probability mass function



Distribution function



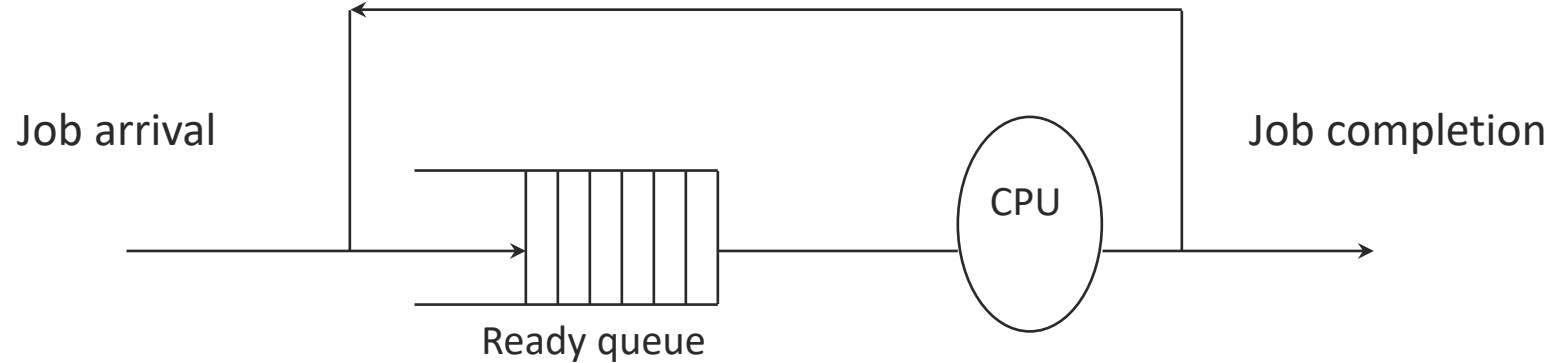
More examples of Bernoulli Trials

Physical situations of interest:

1. Execution of an **if** statement, with success = “**then** clause is executed” and failure = “**else** clause is executed”
2. Examine components produced on an assembly line, with success = “acceptable” and failure = “defective”
3. Transmit binary digits through a communication channel, with success = “digit received correctly” and failure = “digit received incorrectly”

More examples of Bernoulli Trials

4. Consider a time-sharing computer system that allocates a finite quantum (or time slice) to a job scheduled for processor service. Observe n time-slice terminations, with success = “job has completed processing” and failure = “job still requires processing and joins the tail end of the ready queue of processes



Bernoulli Trials (cont.)

- Let 0 denote failure and 1 denote success. Let S_n be the sample space of an experiment involving n Bernoulli trials, defined by:
 - $S_1 = \{0, 1\}$, $S_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$
 - $S_n = \{2^n \text{ n-tuples of } 0\text{'s and } 1\text{'s}\}.$
- The probability assignment over the sample space S_1 is already specified: $p(0) = q$, $p(1) = p$, and $p + q = 1$. We wish to assign probabilities to the points in S_n .

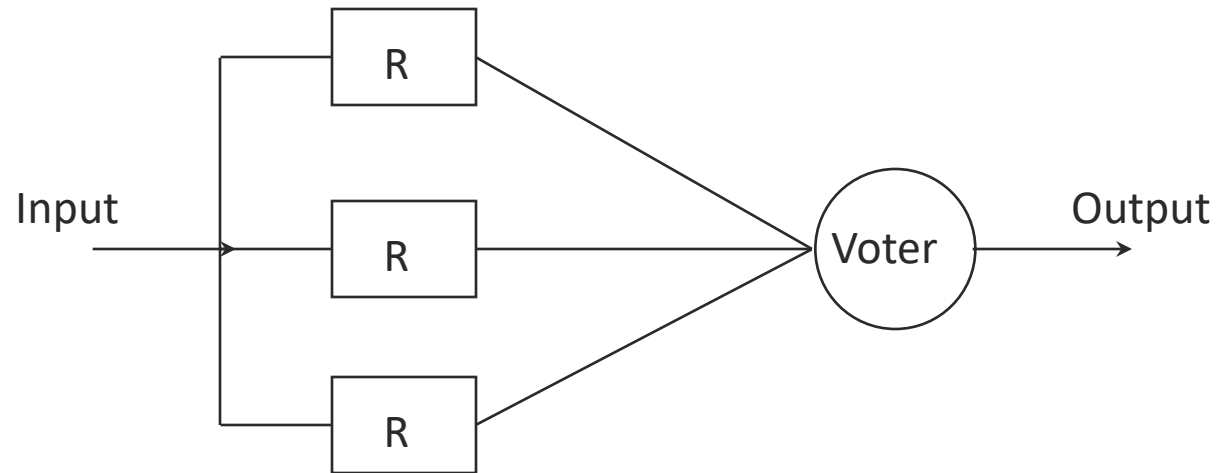
Bernoulli Trials (cont.)

Let A_i = “Success on trial i ” and $\bar{A}_i$ = “Failure on trial i ” then $P(A_i) = p$ and $P(\bar{A}_i) = q$.

Another example of Bernoulli Trials

TMR System Example

Consider a system with triple modular redundancy (TMR). In such a system there are three components, two of which are required to be in working order for the system to function properly (i.e., $n = 3$ and $m = 2$). This is achieved by feeding the outputs of the three components into a majority voter.



Majority Voter Circuit

Truth table of 3-input
voter

INPUT			OUTPUT
A	B	C	$\langle A, B, C \rangle$
0	0	0	0
0	0	1	0
0	1	0	0
1	0	0	0
0	1	1	1
1	0	1	1
1	1	0	1
1	1	1	1

Binomial distribution

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Definition:

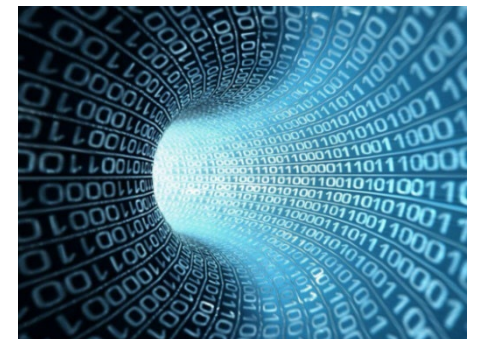
A random variable X has a binomial distribution with parameters n and p if X can take on the values $k = 0, 1, \dots, n$ with probabilities

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Notation: $X \sim \text{Bin}(n, p)$.

A message consists of 10 bits and with probability $p = 0.95$ a bit is transmitted correctly.

What is the probability the message contains exactly 2 errors?



Binomial distribution

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Definition:

A random variable X has a binomial distribution with parameters n and p if X can take on the values $k = 0, 1, \dots, n$ with probabilities

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In general:

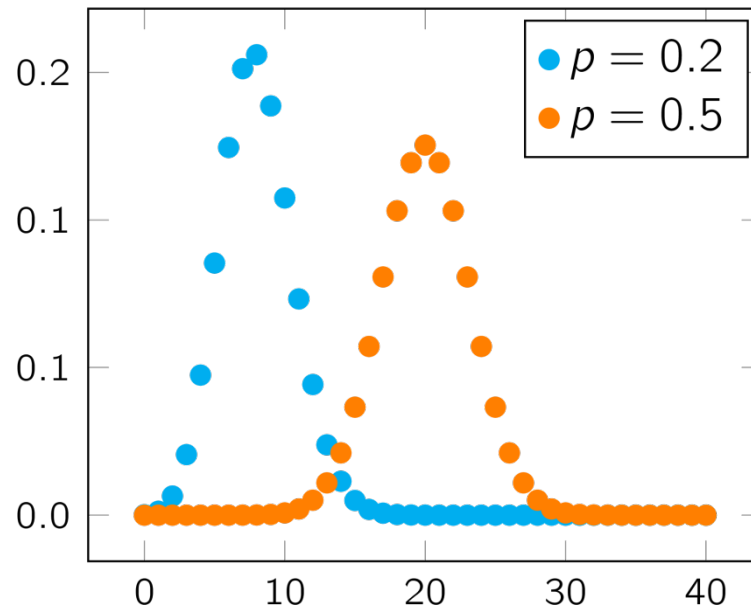
Suppose you conduct n independent experiments where each outcome is either a success or not. With probability p we observe a success. Then we speak of a Binomial distribution with parameters n and p .



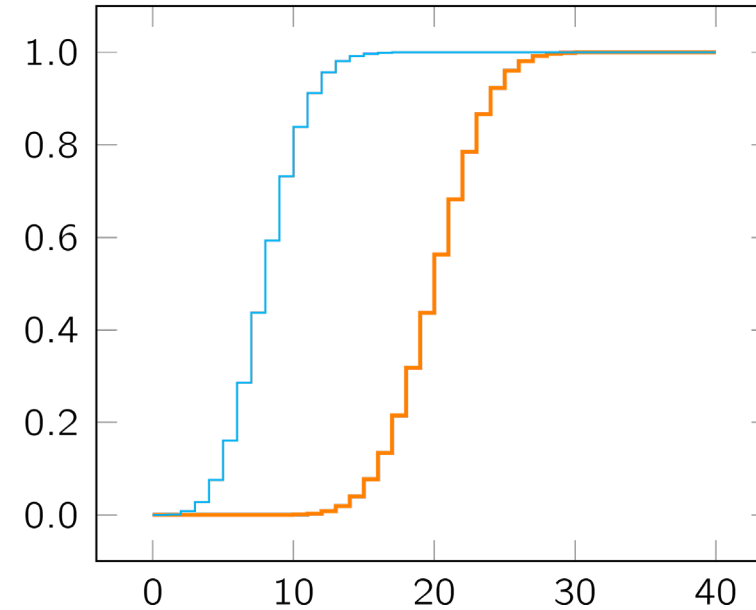
Binomial distribution

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Probability mass function



Distribution function



Binomial Random Variable Example 1

- Four fair coins are flipped. Outcomes are assumed independent, what is the probability that two heads and two tails are obtained?

Binomial Random Variable Example 2

- It is known that **an item produced** by a certain machine will be **defective with probability 0.1**; independent of any other item. What is the probability that in a sample of three items, **at most one will be defective**?

Binomial Random Variable Example 3

- Suppose that an airplane engine will fail, when in flight, with probability $1-p$ independently from engine to engine. Also, suppose that the airplane makes a successful flight if at least 50 percent of its engines remain operative. For what values of p is a four-engine plane preferable to a two-engine plane?

Geometric distribution

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As soon as the source starts transmitting you check whether an erroneous bit is sent.

What is the probability that the fifth bit sent is the first bit with an error?
(an error occurs with probability 0.05)

Definition:

A random variable X has a geometric distribution with parameter p if X can take on the values $k = 1, 2, 3, \dots$, with probabilities

$$P(X = k) = p \cdot (1 - p)^{k-1}.$$

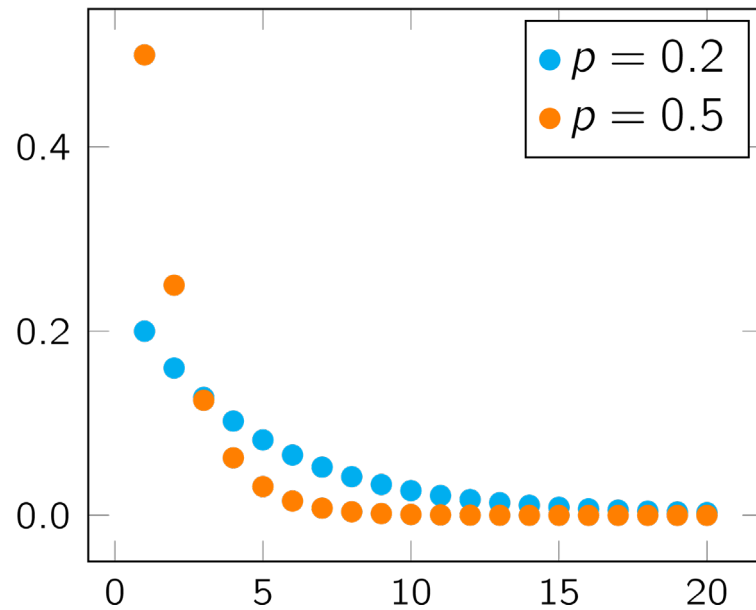
Notation: $X \sim \text{Geo}(p)$.



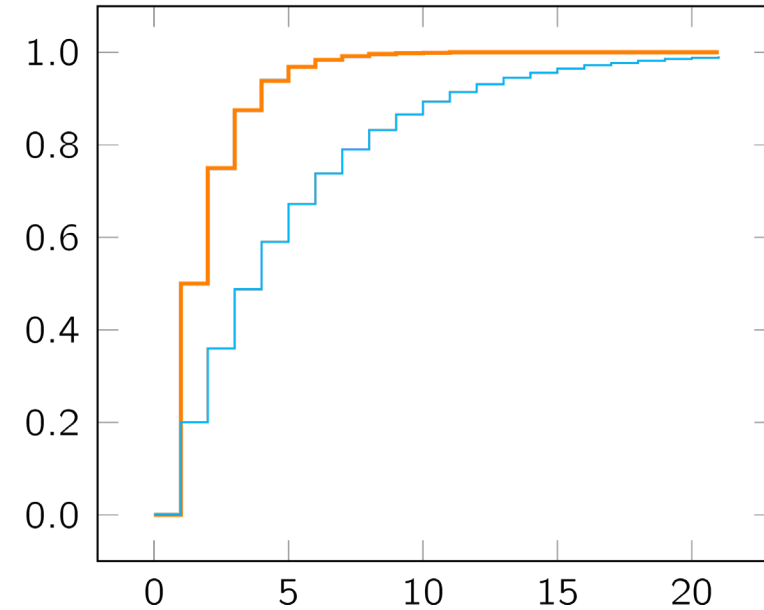
Geometric distribution

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Probability mass function



Distribution function



Geometric Distribution Examples

Consider a **repeat** loop

repeat S until B

The number of tries until B (success) is reached (i.e., includes B), is a geometrically distributed Random Variable with parameter p .

B	repeat #	# of S executions
False	1	1
False	2	2
False	3	3
True	4	4

Discrete distributions

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Shuffle a deck of cards (52) well and place one on the table. Note the result, put the card back in the deck and repeat the process. Do this 10 times. The total number of times you noted a spade is called X .

The distribution of X is of type:

- A. Bernoulli
- B. Binomial
- C. Geometric
- D. Other

Discrete distributions

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Put the cards of a well-shuffled deck (52 cards) one by one on the table until the first ace appears. The number of cards until this first ace is called X . The distribution of X is of type:

- A. Bernoulli
- B. Binomial
- C. Geometric
- D. Other

Discrete distributions

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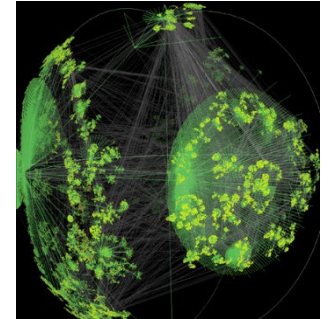
You toss n coins, each showing heads with probability p , independently of the other tosses. Each coin that shows tails is tossed again. Let X be the total number of heads. What type of distribution does X have?

- A. Bernoulli
- B. Binomial
- C. Geometric
- D. Other

Poisson distribution

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What is the probability that your favourite website has 3 hits in a second?



Definition:

A random variable X has a Poisson distribution with parameter μ if X can take on the values $k = 0, 1, 2, \dots$, with probabilities

$$P(X = k) = \frac{\mu^k}{k!} e^{-\mu}.$$

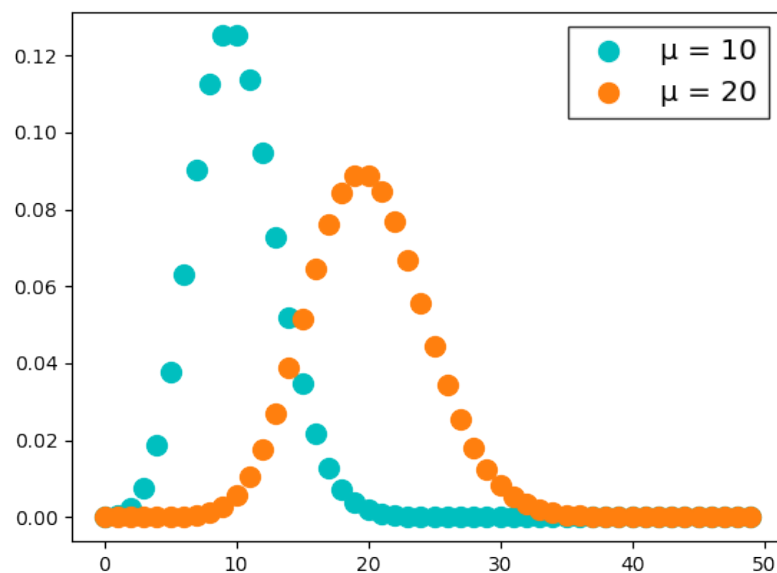
Notation: $X \sim \text{Pois}(\mu)$



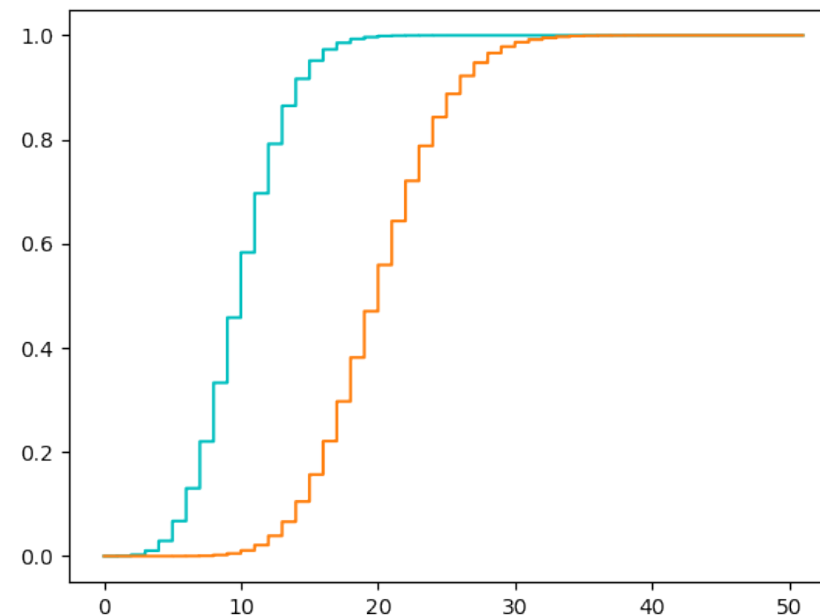
Poisson distribution

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Probability mass function



Distribution function



Practice: Poisson Random Variables Example

A cloud computing system failure occurs according to a Poisson distribution with an average of 3 failures every 10 weeks.

- Calculate the probability that there will not be more than one failure during a particular week
- Calculate the probability that there will be at least one failure during a particular month (4 weeks)

Overview distributions

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Distribution	Parameter(s)	Outcomes	Probability mass function
Ber(p)	$0 \leq p \leq 1$	$\{0, 1\}$	$p(0) = 1 - p$ $p(1) = p$
Bin(n, p)	$n = 1, 2, \dots$ $0 \leq p \leq 1$	$\{0, 1, 2, \dots, n\}$	$p(k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Geo(p)	$0 \leq p \leq 1$	$\{1, 2, 3, \dots\}$	$p(k) = (1 - p)^{k-1} p$
Pois(μ)	$\mu > 0$	$\{0, 1, 2, \dots\}$	$p(k) = \frac{\mu^k}{k!} e^{-\mu}$



Practice now

31

You and a friend want to go to a concert, but unfortunately only one ticket is still available. The man who sells the tickets decides to toss a coin until heads appears. In each toss heads appears with probability p , where $0 < p < 1$, independent of each of the previous tosses. If the number of tosses needed is odd, your friend is allowed to buy the ticket; otherwise you can buy it. Would you agree to this arrangement?



Practice now

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Practice now

31

Let X be a discrete random variable with probability mass function p given by

a	-1	0	1	2
$p(a)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{2}$

and $p(a) = 0$ for all other a .

- Let the random variable Y be defined by $Y = X^2$. Calculate the probability mass function of Y .
- Calculate the value of the distribution functions of X and Y in $a = 1$, $a = 3/4$, and $a = \pi - 3$.



Practice now

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and $p(a) = 0$ for all other a .

- Let the random variable Y be defined by $Y = X^2$. Calculate the probability mass function of Y .
- Calculate the value of the distribution functions of X and Y in $a = 1$, $a = 3/4$, and $a = \pi - 3$.



Practice now

31

Early in the morning, a group of m people decides to use the elevator in an otherwise deserted building of 21 floors. Each of these persons chooses his or her floor independently of the others, and—from our point of view— completely at random, so that each person selects a floor with probability $1/21$. Let S_m be the number of times the elevator stops. In order to study S_m , we introduce for $i = 1, 2, \dots, 21$ random variables R_i , given by

$$R_i = \begin{cases} 1 & \text{if the elevator stops at the } i\text{th floor} \\ 0 & \text{if the elevator does not stop at the } i\text{th floor.} \end{cases}$$

- a) What is the distribution of R_i ?
- b) What's the distribution of S_m ? Derive the probability mass function for S_1 , S_2 and S_3 ?



Practice now

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